Essential Nutrition For Healthy Diet

Name: Mahaiswarya Karuturi
Student ID: 2520030550
Branch: CSE
Cluster- 1
Section- 2

Under the Guidance of
Supervisor Name: T SP Madhuri
Assistant Proffesor
Department of CSE

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- Proposed System
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Abstract

Meal Blender is a smart web-based Nutrition Calculator designed to help users analyze and manage their daily food intake efficiently. The system contains 200 food items categorized into 60 vegetables, 50 fruits, 22 dairy products, and 68 Indian food items. Users can search foods using an advanced search bar and filter items based on categories for quick access.

The application allows multiple food selection and calculates individual as well as total nutritional values including calories, protein, carbohydrates, and fat. It also provides a Daily Goal Tracking feature based on a 2000 calorie diet model with visual progress indicators. Built using HTML, CSS, and JavaScript, the system performs all processing on the client side, making it a standalone and interactive nutrition management solution.

Existing System

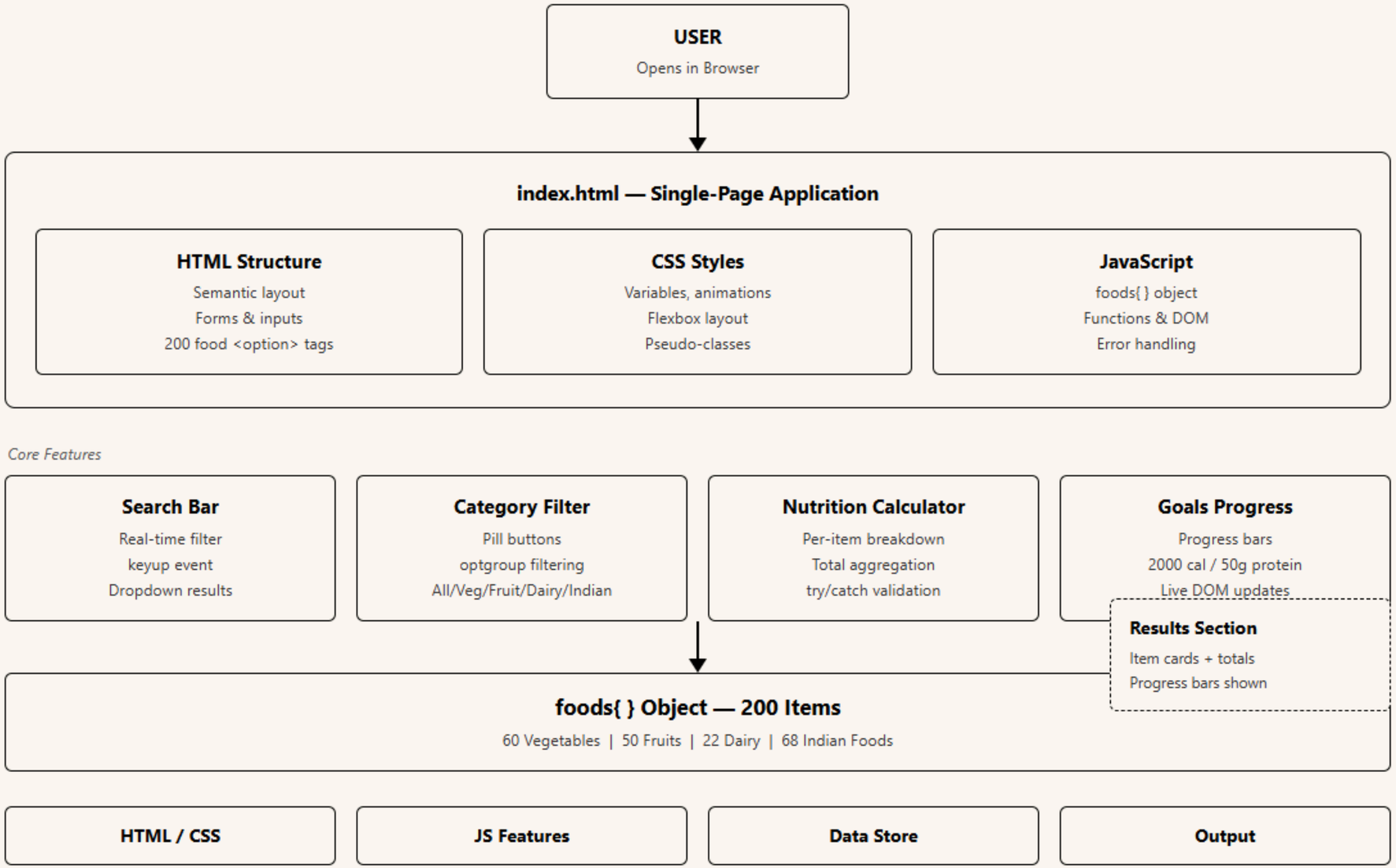
- Most nutrition apps focus on Western food databases
- Limited representation of Indian cuisine and local ingredients
- Require manual entry of portion sizes and calculations
- Complex interfaces not user-friendly for quick meal planning
- Lack of visual feedback on daily nutritional goals
- No integrated search with categorized food groups
- Subscription-based models with premium features locked

Proposed System

- Simple web-based meal builder
- Includes 200 food items database
- Users can add multiple food items
- Instant nutrition calculation
- Search bar for quick food selection
- Category filters for easier browsing
- Daily nutrition goal progress bars
- Clean and user-friendly interface

System Architecture

- User Interface:
  > Built with HTML and CSS
- Interaction Laye:
  > JavaScript handles user actions
- Food Database:
  > Contains 200 food items
- Calculation Engine:
  > Calculates calories, protein, carbs, fat
- Results Display:
  > Shows breakdown and total nutrition


USER
Opens in Browser
index.html — Single-Page Application
HTML Structure
Semantic layout
Forms & inputs
200 food <option> tags
CSS Styles
Variables, animations
Flexbox layout
Pseudo-classes
JavaScript
foods{ } object
Functions & DOM
Error handling
Core Features
Search Bar
Real-time filter
keyup event
Dropdown results
Category Filter
Pill buttons
optgroup filtering
All/Veg/Fruit/Dairy/Indian
Nutrition Calculator
Per-item breakdown
Total aggregation
try/catch validation
Goals Progress
Progress bars
2000 cal / 50g protein
Live DOM updates
Results Section
Item cards + totals
Progress bars shown
foods{ } Object — 200 Items
60 Vegetables | 50 Fruits | 22 Dairy | 68 Indian Foods
HTML / CSS
JS Features
Data Store
Output

Project Modules

- Search Module
- Category Filter Module
- Meal Builder Module
- Nutrition Calculation Module
- Results Display Module
- Goal Tracking Module

Objectives

- Build a simple nutrition calculator
- Help users understand food nutrition
- Provide 200 food items database
- Enable interactive meal building
- Display accurate nutrition breakdown
- Improve user-friendly web interface

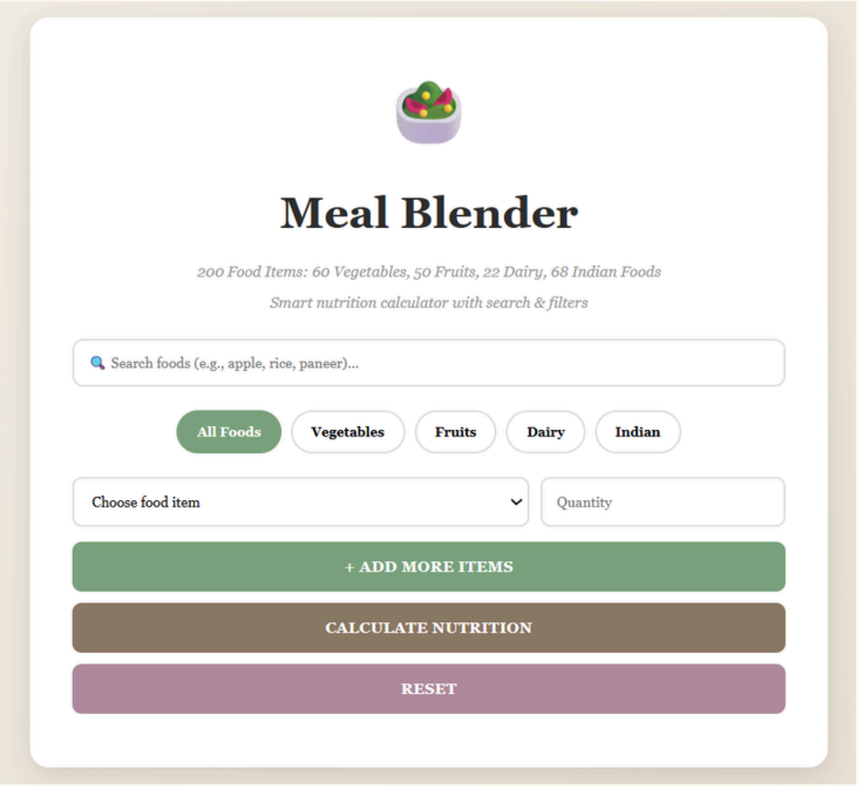
Co’s Attainments – FWD

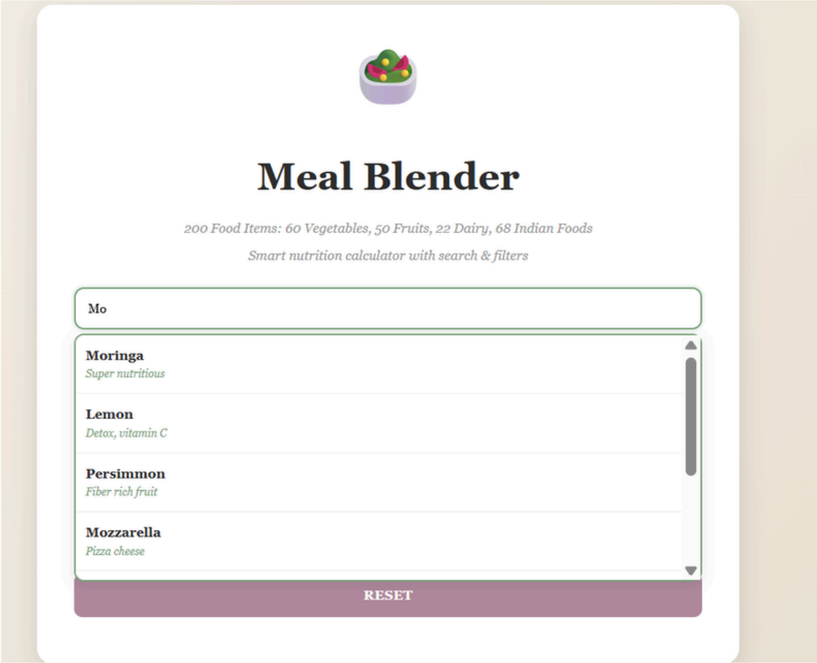
- CO1 – HTML structure, CSS styling
- CO2 – Forms, Flexbox
- CO3 – Objects, Functions, Loops
- CO4 – DOM Manipulation
- CO5 – Try-Catch, Validation

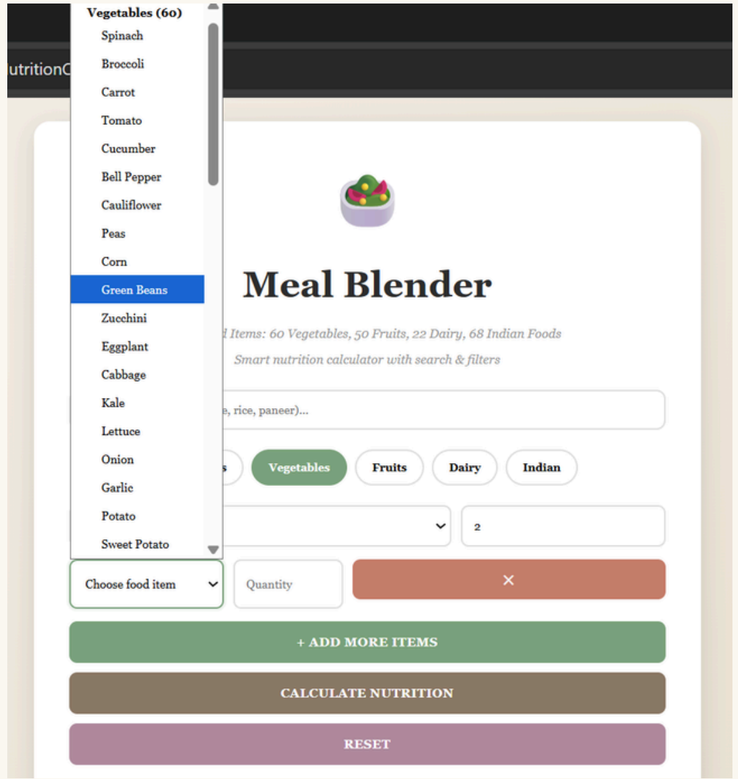
Co’s Attainments – DSA

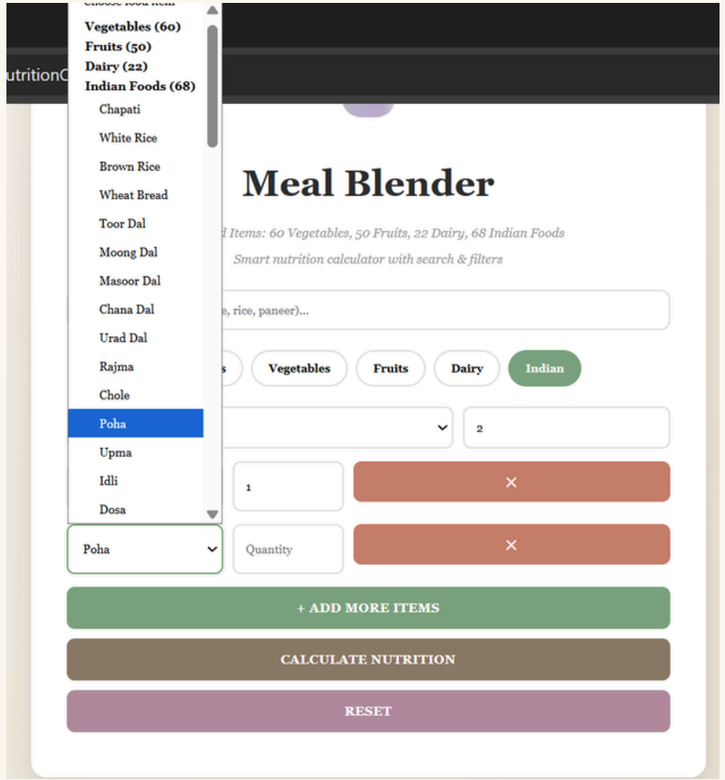
- CO1 – Algorithm analysis & sorting/searching
- CO2 – ADTs using arrays & linked lists
- CO3 – Stacks, Queues, Heaps
- CO4 – Hash tables & Java Collections

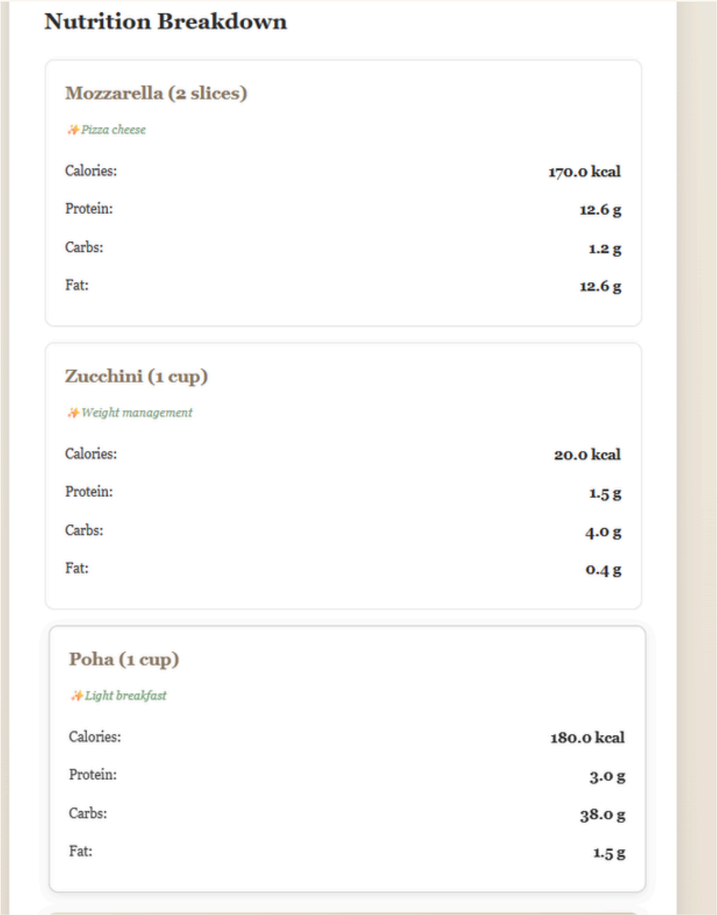
FWD

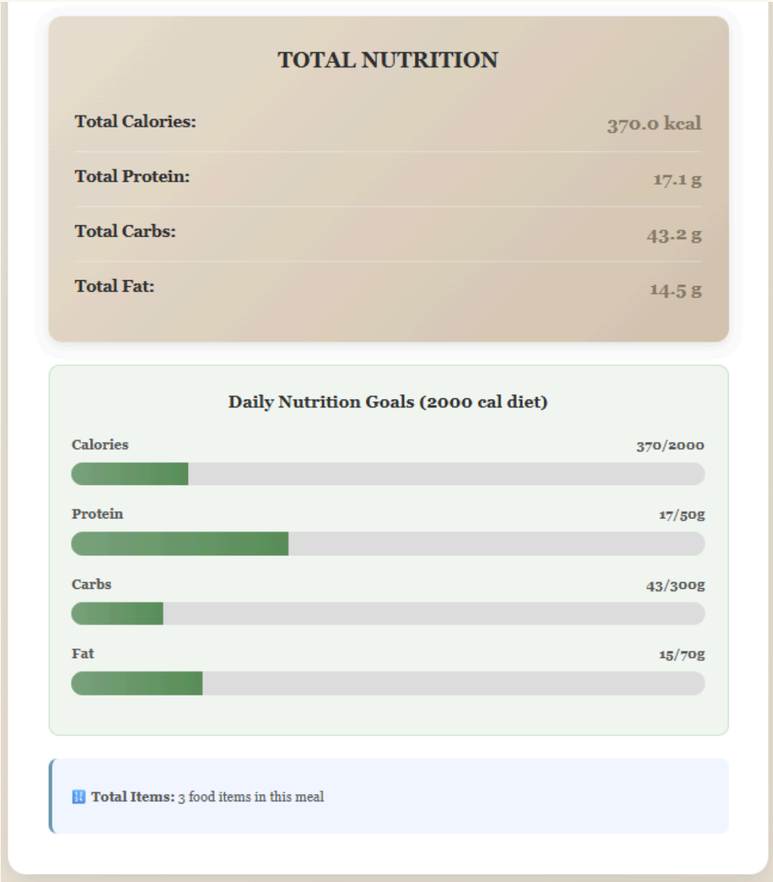

Meal Blender
200 Food Items: 60 Vegetables, 50 Fruits, 22 Dairy, 68 Indian Foods
Smart nutrition calculator with search & filters
Search foods (e.g., apple, rice, paneer)...
All Foods
Vegetables
Fruits
Dairy
Indian
Choose food item
Quantity
+ ADD MORE ITEMS
CALCULATE NUTRITION
RESET


Meal Blender
200 Food Items: 60 Vegetables, 50 Fruits, 22 Dairy, 68 Indian Foods
Smart nutrition calculator with search & filters
Mo
Moringa
Super nutritious
Lemon
Detox, vitamin C
Persimmon
Fiber rich fruit
Mozzarella
Pizza cheese
RESET


Vegetables (60)
Spinach
Broccoli
Carrot
Tomato
Cucumber
Bell Pepper
Cauliflower
Peas
Corn
Green Beans
Zucchini
Eggplant
Cabbage
Kale
Lettuce
Onion
Garlic
Potato
Sweet Potato
Meal Blender
Vegetables
Fruits
Dairy
Indian
2
Choose food item
Quantity
+ ADD MORE ITEMS
CALCULATE NUTRITION
RESET


Vegetables (60)
Fruits (50)
Dairy (22)
Indian Foods (68)
Chapati
White Rice
Brown Rice
Wheat Bread
Toor Dal
Moong Dal
Masoor Dal
Chana Dal
Urad Dal
Rajma
Chole
Poha
Upma
Idli
Dosa
Meal Blender
Smart nutrition calculator with search & filters
Vegetables
Fruits
Dairy
Indian
Quantity
Poha
+ ADD MORE ITEMS
CALCULATE NUTRITION
RESET


Nutrition Breakdown
Mozzarella (2 slices)
Pizza cheese
Calories: 170.0 kcal
Protein: 12.6 g
Carbs: 1.2 g
Fat: 12.6 g
Zucchini (1 cup)
Weight management
Calories: 20.0 kcal
Protein: 1.5 g
Carbs: 4.0 g
Fat: 0.4 g
Poha (1 cup)
Light breakfast
Calories: 180.0 kcal
Protein: 3.0 g
Carbs: 38.0 g
Fat: 1.5 g


TOTAL NUTRITION
Total Calories: 370.0 kcal
Total Protein: 17.1 g
Total Carbs: 43.2 g
Total Fat: 14.5 g
Daily Nutrition Goals (2000 cal diet)
Calories 370/2000
Protein 17/50g
Carbs 43/300g
Fat 15/70g
Total Items: 3 food items in this meal

DSA

```
PS C:\DSA Project>  & 'C:\Program Files\Java\jdk-25\bin\java.exe' '--enable-preview' '-XX:+ShowCode
DetailsInExceptionMessages' '-cp' 'C:\Users\mahai\AppData\Roaming\Code\User\workspaceStorage\80ab5d
9b153d26eec588a3bc7caf2883\redhat.java\jdt_ws\DSA Project_433670b3\bin' 'MealBlender'
======================================================================
   MEAL BLENDER - Nutrition Calculator
   200 Foods: Vegetables | Fruits | Dairy | Indian
   DS Concepts: HashMap | LinkedList | Stack | Queue | Heap
======================================================================

  MAIN MENU
  1. Search food items              [CO1: Linear Search O(n)]
  2. Browse by category             [CO4: HashMap + Selection Sort]
  3. View current meal              [CO2: LinkedList traverse]
  4. Calculate nutrition            [CO5: Goal tracking]
  5. Undo last add                  [CO3: Stack LIFO]
  6. Top low-calorie foods          [CO3: Min-Heap]
  7. Top high-protein foods         [CO1: Merge Sort O(n log n)]
  8. Binary search food name        [CO1: Binary Search O(log n)]
  9. Recent searches                [CO4: Deque]
  10. Clear meal
  0. Exit

  Your choice: 1

  Search food name: Spinach

  Search results for "Spinach" (1 found):
  ----------------------------------------------------------------
    1. [veg   ] Spinach                  | Cal:    7.0 | Prot:   0.9g | Carbs:   1.1g | Fat:   0.1
g
```

```
Your choice: 2

Categories: veg | fruit | dairy | indian
Enter category: fruit

FRUIT foods (50):
----------------------------------------------------------------
  1. Amla                    | Cal:   44.0 | Prot:   0.9g | Carbs:  10.2g | Fat:   0.6g
  2. Apple                   | Cal:   95.0 | Prot:   0.5g | Carbs:  25.0g | Fat:   0.3g
  3. Apricot                 | Cal:   17.0 | Prot:   0.5g | Carbs:   4.0g | Fat:   0.1g
  4. Avocado                 | Cal:  234.0 | Prot:   2.9g | Carbs:  12.0g | Fat:  21.0g
  5. Banana                  | Cal:  105.0 | Prot:   1.3g | Carbs:  27.0g | Fat:   0.4g
  6. Blackberry              | Cal:    0.4 | Prot:   0.0g | Carbs:   0.1g | Fat:   0.0g
  7. Blueberry               | Cal:    0.6 | Prot:   0.0g | Carbs:   0.1g | Fat:   0.0g
  8. Cantaloupe              | Cal:   60.0 | Prot:   1.5g | Carbs:  14.4g | Fat:   0.3g
  9. Cherry                  | Cal:    0.6 | Prot:   0.0g | Carbs:   0.2g | Fat:   0.0g
 10. Coconut                 | Cal:  283.0 | Prot:   2.7g | Carbs:  12.0g | Fat:  27.0g
 11. Cranberry               | Cal:    0.5 | Prot:   0.0g | Carbs:   0.1g | Fat:   0.0g
 12. Custard Apple           | Cal:  236.0 | Prot:   5.0g | Carbs:  59.0g | Fat:   0.6g
 13. Date                    | Cal:   20.0 | Prot:   0.2g | Carbs:   5.0g | Fat:   0.0g
 14. Dragon Fruit            | Cal:  102.0 | Prot:   2.0g | Carbs:  22.0g | Fat:   0.6g
 15. Fig                     | Cal:   47.0 | Prot:   0.5g | Carbs:  12.0g | Fat:   0.2g
 16. Gooseberry              | Cal:   66.0 | Prot:   1.3g | Carbs:  15.0g | Fat:   0.9g
 17. Grapefruit              | Cal:   52.0 | Prot:   0.9g | Carbs:  13.0g | Fat:   0.2g
 18. Grapes                  | Cal:    0.7 | Prot:   0.0g | Carbs:   0.2g | Fat:   0.0g
 19. Guava                   | Cal:   37.0 | Prot:   1.4g | Carbs:   8.0g | Fat:   0.5g
 20. Honeydew                | Cal:   64.0 | Prot:   1.0g | Carbs:  16.0g | Fat:   0.2g
 21. Jackfruit               | Cal:  155.0 | Prot:   2.8g | Carbs:  38.0g | Fat:   0.5g
 22. Jamun                   | Cal:    0.6 | Prot:   0.0g | Carbs:   0.1g | Fat:   0.0g
 23. Kiwi                    | Cal:   42.0 | Prot:   0.8g | Carbs:  10.0g | Fat:   0.4g
 24. Lemon                   | Cal:   17.0 | Prot:   0.6g | Carbs:   5.0g | Fat:   0.2g
 25. Lime                    | Cal:   20.0 | Prot:   0.5g | Carbs:   7.0g | Fat:   0.1g
 26. Lychee                  | Cal:    0.7 | Prot:   0.0g | Carbs:   0.2g | Fat:   0.0g
 27. Mango                   | Cal:  202.0 | Prot:   2.8g | Carbs:  50.0g | Fat:   1.3g
 28. Mangosteen              | Cal:   73.0 | Prot:   0.4g | Carbs:  17.9g | Fat:   0.6g
 29. Melon                   | Cal:   54.0 | Prot:   1.3g | Carbs:  13.0g | Fat:   0.3g
 30. Mulberry                | Cal:   60.0 | Prot:   2.0g | Carbs:  14.0g | Fat:   0.5g
 31. Nectarine               | Cal:   63.0 | Prot:   1.5g | Carbs:  15.1g | Fat:   0.5g
 32. Orange                  | Cal:   62.0 | Prot:   1.2g | Carbs:  15.0g | Fat:   0.2g
 33. Papaya                  | Cal:   62.0 | Prot:   0.7g | Carbs:  16.0g | Fat:   0.4g
 34. Passion Fruit           | Cal:   17.0 | Prot:   0.4g | Carbs:   4.0g | Fat:   0.1g
 35. Peach                   | Cal:   58.0 | Prot:   1.4g | Carbs:  14.0g | Fat:   0.4g
 36. Pear                    | Cal:  101.0 | Prot:   0.6g | Carbs:  27.0g | Fat:   0.2g
 37. Persimmon               | Cal:  118.0 | Prot:   1.0g | Carbs:  31.0g | Fat:   0.3g
 38. Pineapple               | Cal:   82.0 | Prot:   0.9g | Carbs:  22.0g | Fat:   0.2g
 39. Plum                    | Cal:   30.0 | Prot:   0.5g | Carbs:   8.0g | Fat:   0.2g
 40. Pomegranate             | Cal:  144.0 | Prot:   3.0g | Carbs:  32.0g | Fat:   2.0g
 41. Pomelo                  | Cal:   72.0 | Prot:   1.4g | Carbs:  18.0g | Fat:   0.1g
 42. Rambutan                | Cal:    7.0 | Prot:   0.1g | Carbs:   1.6g | Fat:   0.0g
 43. Raspberry               | Cal:    0.5 | Prot:   0.0g | Carbs:   0.1g | Fat:   0.0g
 44. Sapodilla               | Cal:   83.0 | Prot:   0.4g | Carbs:  20.0g | Fat:   1.1g
 45. Star Fruit              | Cal:   28.0 | Prot:   1.0g | Carbs:   6.0g | Fat:   0.3g
 46. Strawberry              | Cal:    0.3 | Prot:   0.0g | Carbs:   0.1g | Fat:   0.0g
 47. Sweet Lime              | Cal:   43.0 | Prot:   0.7g | Carbs:   9.3g | Fat:   0.3g
 48. Tamarind                | Cal:  287.0 | Prot:   3.4g | Carbs:  75.0g | Fat:   0.7g
```

DSA

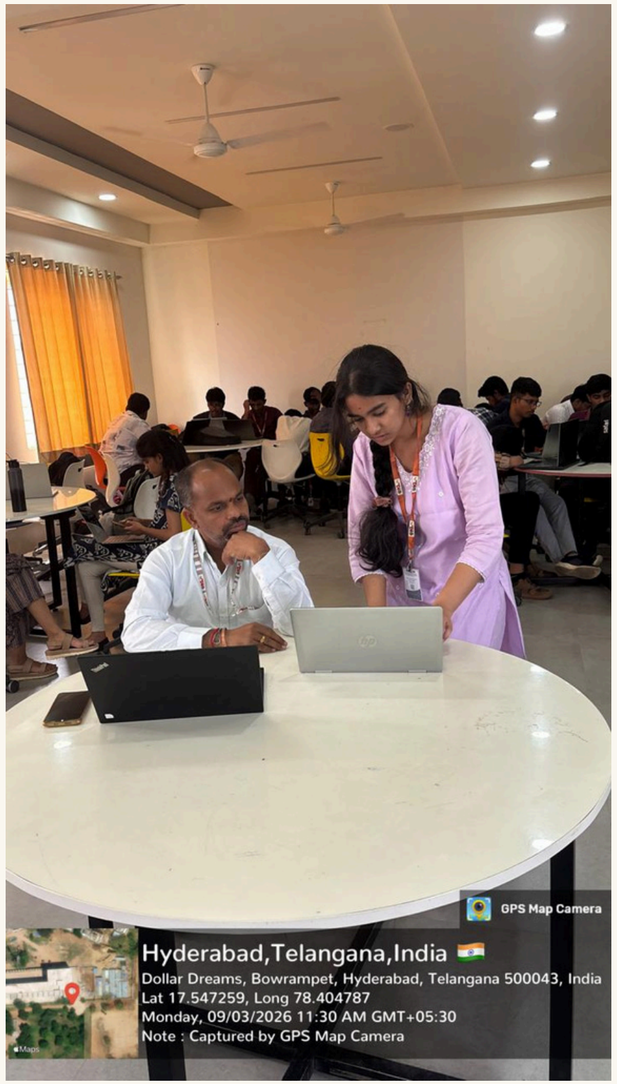
```
 50. Watermelon                | Cal:   46.0 | Prot:   0.9g | Carbs:  12.0g | Fat:   0.2g

Enter number to add to meal (0 to skip): 13
Enter quantity: 2
Added: Date

MAIN MENU
1. Search food items             [CO1: Linear Search O(n)]
2. Browse by category            [CO4: HashMap + Selection Sort]
3. View current meal             [CO2: LinkedList traverse]
4. Calculate nutrition           [CO5: Goal tracking]
5. Undo last add                 [CO3: Stack LIFO]
6. Top low-calorie foods         [CO3: Min-Heap]
7. Top high-protein foods        [CO1: Merge Sort O(n log n)]
8. Binary search food name       [CO1: Binary Search O(log n)]
9. Recent searches               [CO4: Deque]
10. Clear meal
0. Exit

Your choice: 3


CURRENT MEAL ITEMS (2):
-------------------------------------------------------
 1. Spinach                    Qty: 3.0 cup
 2. Date                       Qty: 2.0 date

MAIN MENU
1. Search food items             [CO1: Linear Search O(n)]
2. Browse by category            [CO4: HashMap + Selection Sort]
3. View current meal             [CO2: LinkedList traverse]
4. Calculate nutrition           [CO5: Goal tracking]
5. Undo last add                 [CO3: Stack LIFO]
6. Top low-calorie foods         [CO3: Min-Heap]
7. Top high-protein foods        [CO1: Merge Sort O(n log n)]
8. Binary search food name       [CO1: Binary Search O(log n)]
9. Recent searches               [CO4: Deque]
10. Clear meal
0. Exit
```

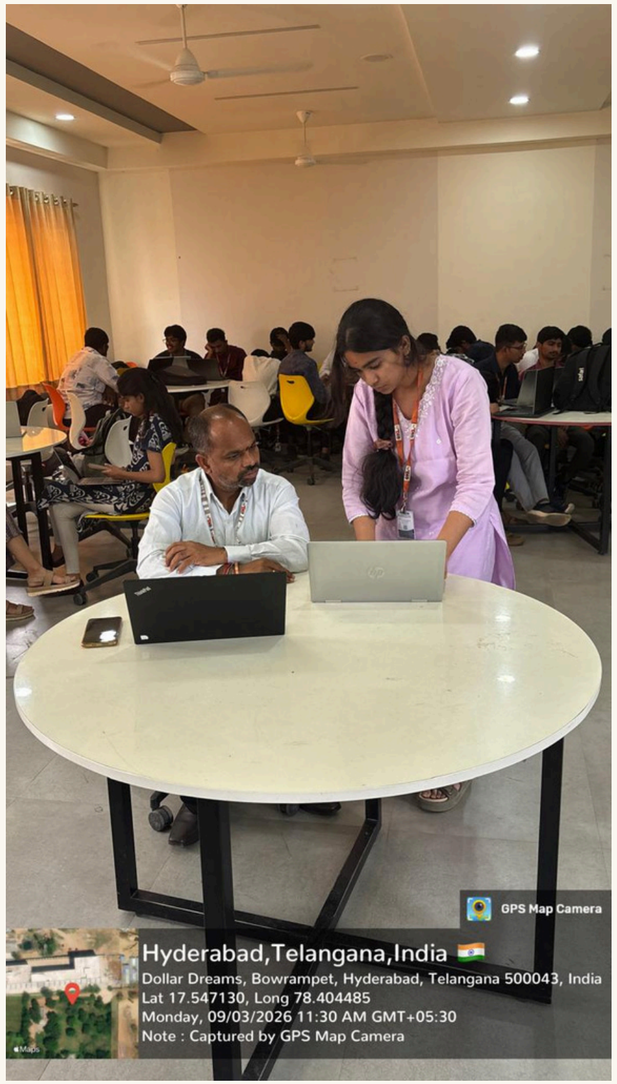
```
  Your choice: 4


  ======================================================================
  NUTRITION BREAKDOWN
  ======================================================================

  Spinach                    (3.0 cup)
  Benefit:         Iron rich, immunity boost
  Calories:    21.0 kcal  |  Protein:    2.7 g
  Carbs:        3.3 g     |  Fat:        0.3 g
  -------------------------------------------------------

  Date                       (2.0 date)
  Benefit:         Energy booster
  Calories:    40.0 kcal  |  Protein:    0.4 g
  Carbs:       10.0 g     |  Fat:        0.0 g
  -------------------------------------------------------

  TOTAL NUTRITION (2 items)
  =======================================================
  Total Calories :     61.0 kcal
  Total Protein  :      3.1 g
  Total Carbs    :     13.3 g
  Total Fat      :      0.3 g

  DAILY GOALS (2000 kcal diet)
  -------------------------------------------------------
  Calories:  [--------------------]  61.0/2000kcal (3%)
  Protein:   [#-------------------]   3.1/50g (6%)
  Carbs:     [--------------------]  13.3/300g (4%)
  Fat:       [--------------------]   0.3/70g (0%)
  ======================================================================
```

Conclusion

- Developed a web-based nutrition calculator
- Supports 200 food items
- Allows custom meal creation
- Provides instant nutrition breakdown
- Includes search, filter, and goal tracking
- Demonstrates frontend and JavaScript concepts
- Useful for healthy diet planning

Geo- Tag Photos





THANK YOU